Assignment1

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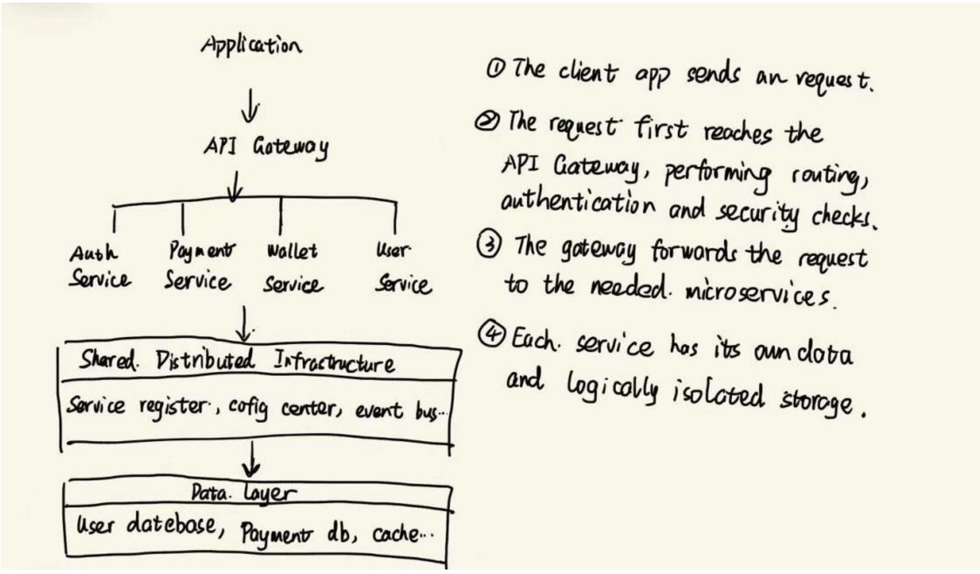
Question1

A large-scale Chinese application, such as Wechat and Alipay, can be understood as a microservices-based distribution system. The platform splits the system into many small services. Each service focuses on one business capability, such as login, payment, search and so on. This design improves scalability, fault isolation of the application.



Question2

1. Application layer

At the application layer, my browser sends a request, such as an HTTPS/HTTP GET, for the video file. The Google server responds with the requested video data.

2. Transport layer

TCP establishes a connection, splits the video data into segments, numbers them, and ensures reliable delivery. If a segment is lost, TCP retransmits it. Port numbers are used so that the data reaches the correct application on my PC.

3. Internet layer

At the internet layer, each TCP segment is placed inside an IP packet. The packet contains the source IP address of the Google server and the destination IP address of my PC. Routers on the Internet examine the IP header and forward the packet toward the destination.

4. Network access layer
Each IP packet is placed into a frame suitable for the local link technology, such as Ethenet in the data centerm fiber links in the ISP backbone, and Wi-Fi in my home network. The data is transmitted as bits over the cables, fiber or radio waves.

Question3

request size : 200 bytes = 1600 bits

respond. size : 5000 bytes = 40000 bits

Data transfer rate: 10Mbps = 10^7 bit/s

request time: $1600 \div 10^7 = 1.6 \times 10^{-4}$ s = 0.16 ms

5ms + 5ms + 0.16 ms = 10.16 ms is request time.

MTU = 1000 bps, $5000 \div 1000 = 5$ packet

1000 bytes × 8 bits / byte $\div 10^7$ bits/s = 0.8 ms

5ms + 5 ms = 10 ms , 10 ms + 0.8ms = 10.8ms

5 × 10.8 ms = 54ms is respond time.

(i) For UDP

10.16ms + 2ms + 54ms = 66.16 ms

(ii) For. TCP

66.16ms + 5ms = 71.16ms.

(iii) For server and client are in the same machine.

5000 bytes × 8 bits/byte $\div 10^7$ bits/second = 4ms

2 × 10 + 0.16 + 4 + 2 = 26.16 ms

Question4

For an IPv6 LAN, the most suitable subnet size is typically /64, such as 2001:db8:1234:0001::/64. It is easy to support 1100 devices and it leaves a very large number of additional subnets for the university.
The university gate: 64-48=16. The university can create 2^{16} subnets.
128-64=64. So the students hall provides 2^{64} usable host addresses.